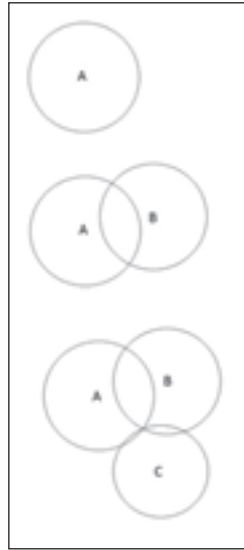


Unites States of America. They orbit at 19,300 km within a time period of 12 hours. The orbits are arranged in a specific way so that at least four satellites are in the line of sight of the receiver.

For the GPS receiver to receive a signal it should have a clear view of the sky. To understand how GPS calculates your exact location, we must first understand Trilateration. Imagine you know your distance from point A, then you know that you can lie anywhere on a circle centred at A and the radius being your distance from the point. Now suppose you know your distance from another point B; you can infer that you'll lie on one of the two intersection points of the two circles. If you know your distance from a third point, you can pinpoint your location as the intersection point of the three circles. This is called 2-D Trilateration. If you extend this logic to 3-D, you'll need your distance from four points. Since you already know your distance from the center of the earth, you need to know your distance from three other points. This information is provided by the GPS satellites.



Trilateration

Every satellite continually transmits messages that include the time at which it was transmitted, its orbital information, its general health and rough orbits of other satellites. When this signal is received by your GPS receiver, there's a very tiny time lag because of the time taken by the signal to reach the GPS receiver. This lag is used to calculate the distance of the satellite (speed of light x time lag). Since the speed of light is very high, the time lag is very low and to measure this lag accurately the clocks on the satellite as well as the GPS receiver should be synchronised down to the nanosecond. Such high level of accuracy is possible using atomic clocks which are very expensive. Instead of having atomic clocks on the satellite and the receiver, only the satellite has the atomic clock and the receiver contains a simple quartz clock. The receiver calculates its own accuracy by receiving signals for four satellites and corrects itself. Every GPS has stored data of where every satellite should be at what time and using that it can calculate the location. Speed is simply calculated by the change in location divided by time.